

Powering On-Prem GenAI From Infrastructure to Impact

Instructors

Sergio Valmorisco
GenAI Apps Squad Lead

Pablo Gómez-Caldito
GenAI Platform Engineer



About us



Sergio Valmorisco



2023-Present
GenAI Apps Squad Lead



2020-2023
Sr. Data Scientist



2020
Data Scientist



2017-2019
Data Analyst



2016-2017
Software Engineer

Bachelor's and Master's
**Telecommunications Engineering,
Computer Science, Data Science**



Pablo Gómez-Caldito



2024-Present
GenAI Platform Engineer



2021-2024
Sr. DevOps Engineer



2020-2021
Platform Engineer



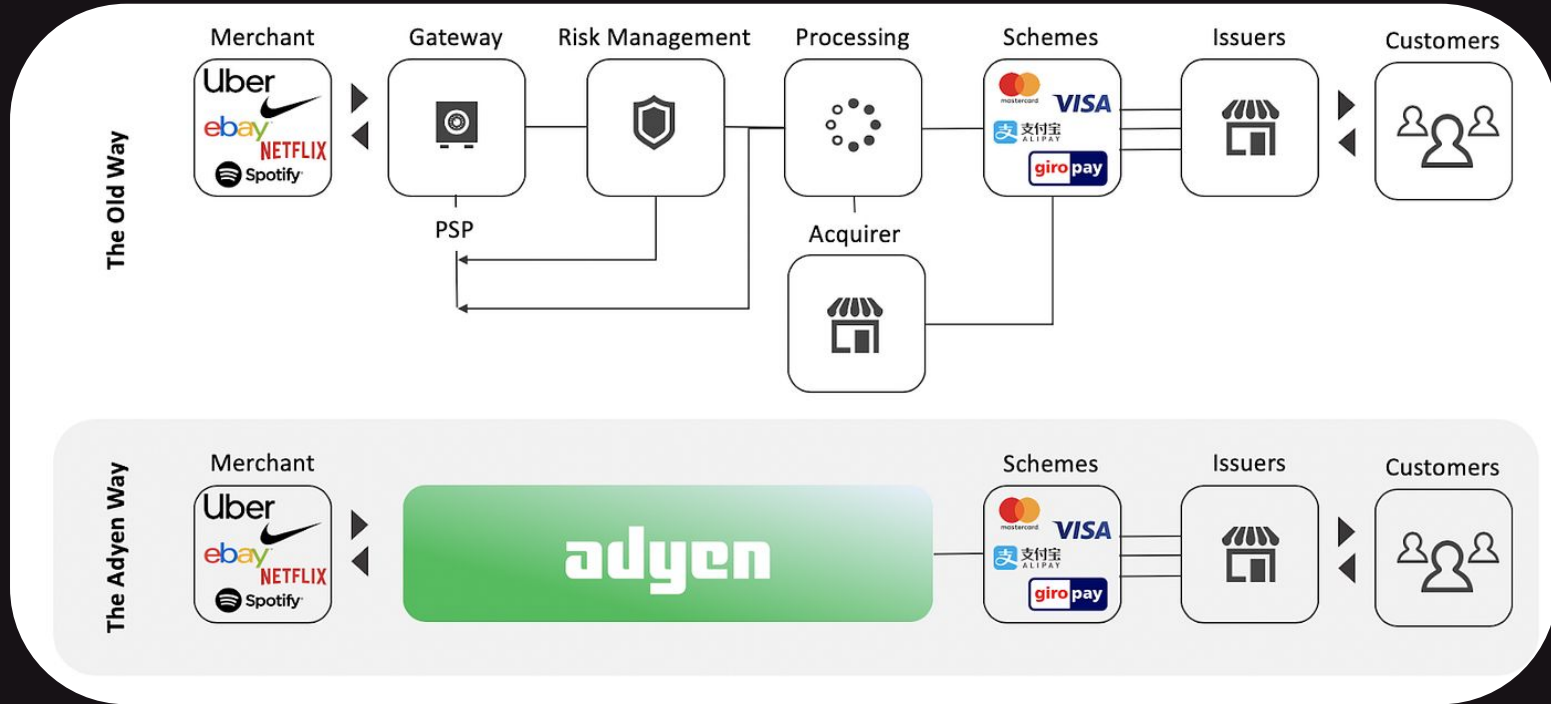
2019-2020
DevOps Intern

Bachelor's in Computer Science
and Engineering

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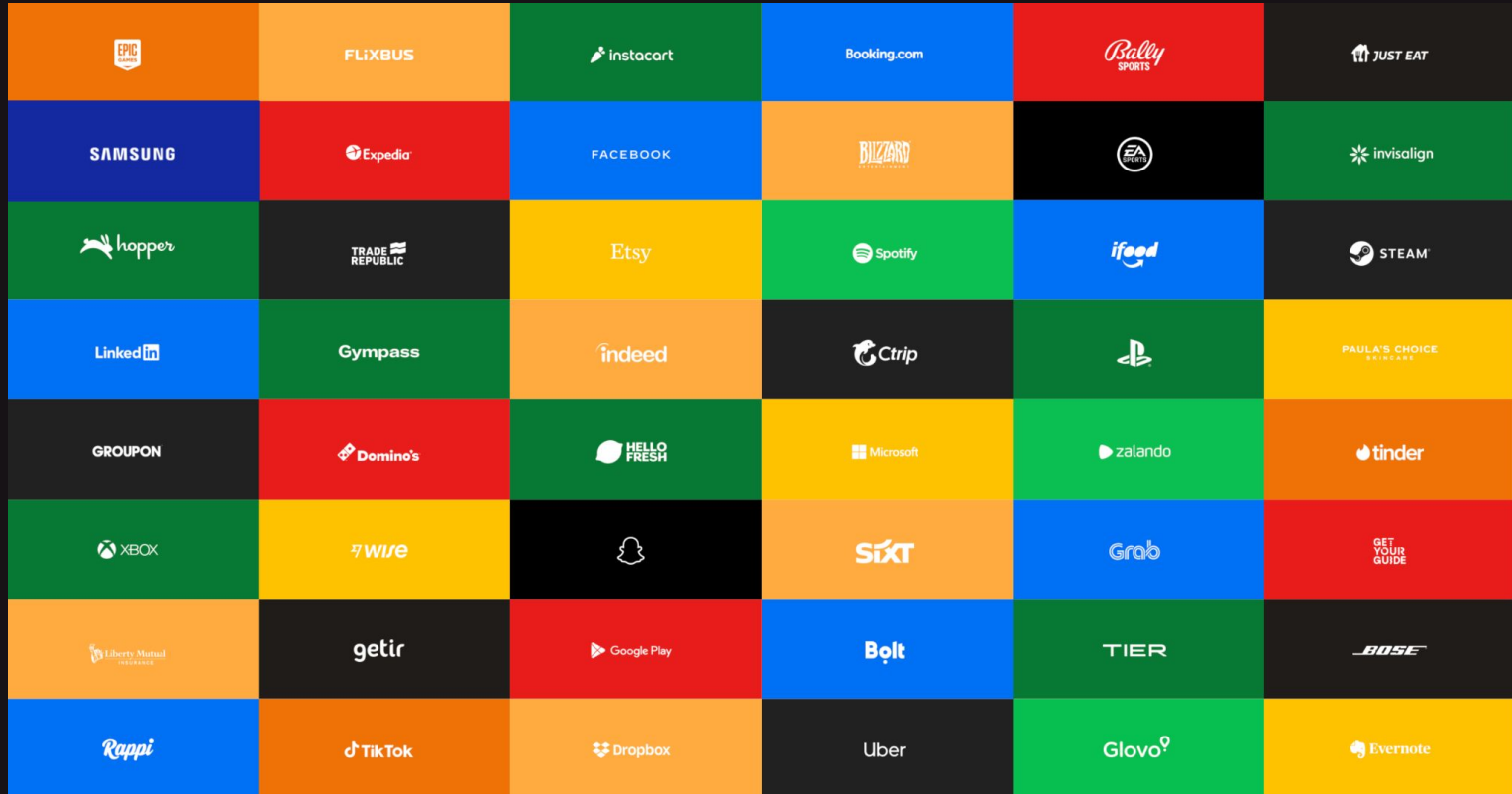
- What does Adyen do?
- GenAI @ Adyen
- GenAI Platform
- GenAI Applications
- Hands-on notebook

What does Adyen do?



In the industry, roughly 15% of online and 5% of real-world transactions are declined. Adyen aims to solve that.

Our customers



GenAI @ **adyen**

Team Lead



Rodrigo Toscano

Product Managers



Mindy Kasting



Elen Gabrielyan



Vanessa Wong

Product Design



Zahra Kazmi

GenAI Applications



Koen Roelofs



Hanna van der Vliet



Antonio Ramos



Jeet Chugh



Sergio Valmorisco



Jorge Zapatero



Zorana Bankovic

GenAI Platform



Pablo Gómez-Caldito



Martín Iglesias



Hector Chamorro



Friso Kingma



Andres Carrillo

10x goal: working 10 times smarter



Generative AI Applications

*Scaling up operations for
specific GenAI business
use-cases.*

Generative AI Platform

*Providing the core
components of the in-house
GenAI Platform.*



GenAI Platform - Why in-house?



Safety and privacy of
sensitive data



Fostering
innovation and
expertise

GenAI Platform - In-house tradeoffs

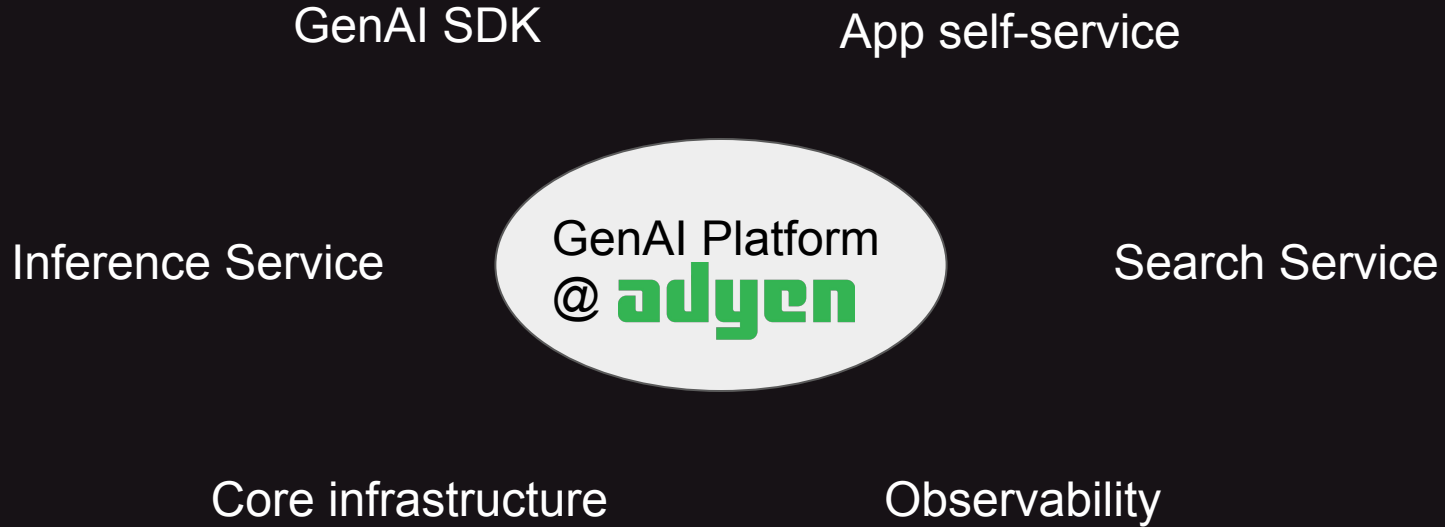


Capacity planning



Deep expertise
required

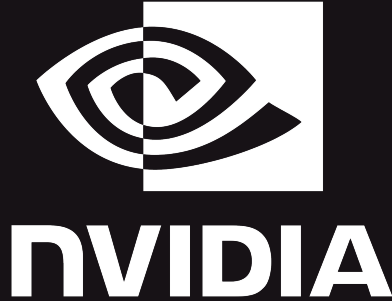
GenAI Platform - Components



GenAI Platform - Core infrastructure



Kubernetes



GPUs and GPU
operator



GitOps with
ArgoCD

GenAI Platform - GenAI SDK



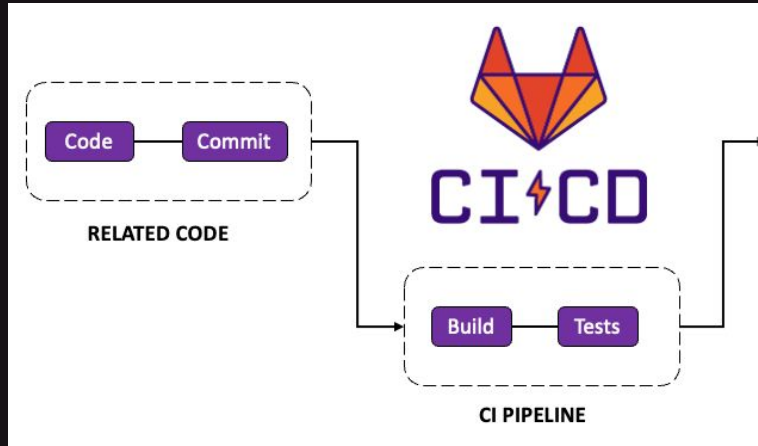
LlamaIndex



LangChain



GenAI Platform - GenAI applications self-service



CD pipelines in
Argo CD



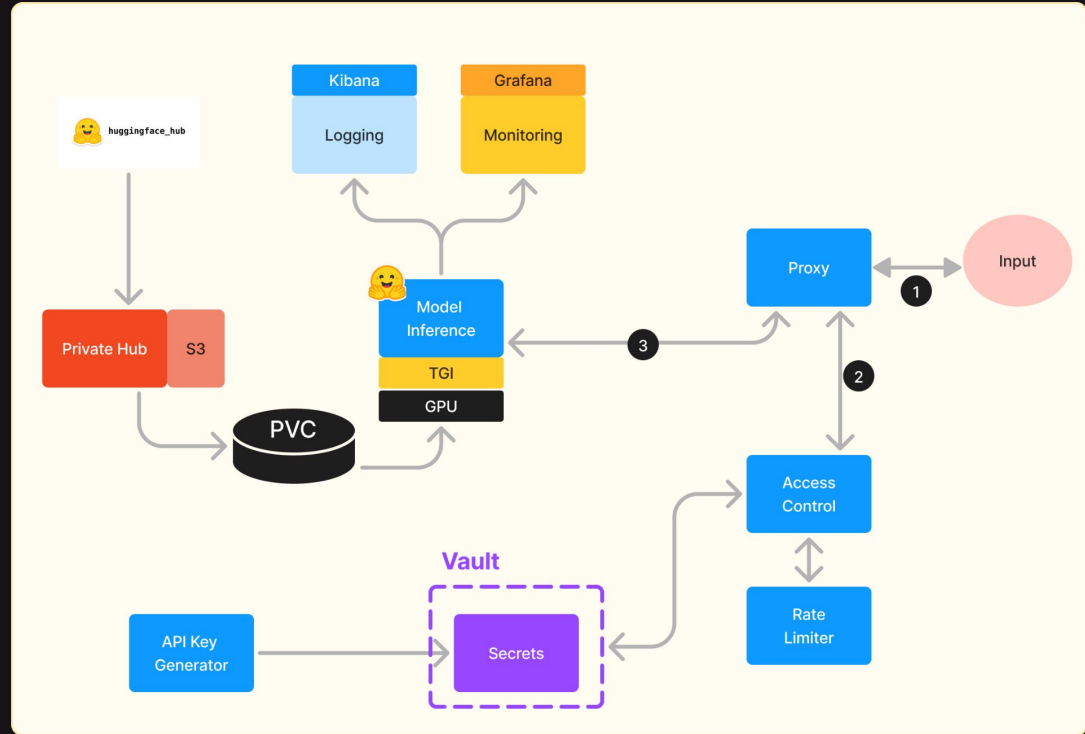
GenAI Platform - Inference service



Using Hugging Face's:

- [Text Generation Inference](#)
- [Text Embeddings Inference](#)
 - (Also rerankers)

LLM Inference Architecture (Per-model)



Link

In figure 3, the first 10 requests smoothly go through the prefill and decode steps, and the TTB is updated accordingly. After this, there are 10 requests in the queue and 10 requests currently decoding, each holding some budget from TTB until they reach their max_new_tokens or generate an EOS token.

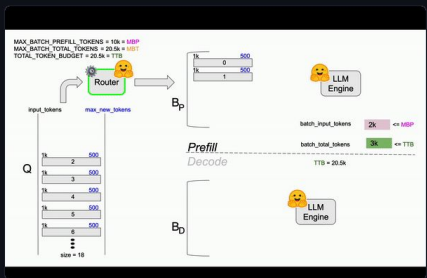
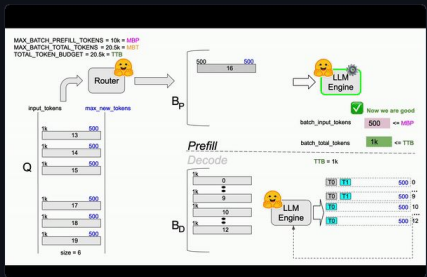


Figure 3: TGI Continuous Batching animation based on TGI router code

We encounter a scenario where requests 13th, 14th, and 15th would exceed the available token budget, preventing them from undergoing the prefill step. As you can see in figure 4, the 16th request, with a smaller token count, fits within the τ_{TB} and successfully prefills the cache, joining the running decoding batch. At this point, the token budget is fully utilized, and we must wait for currently running requests to complete.



 Hugging Face

 Models

 Datasets

 Spaces



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LLM Inference at scale with TGI

 Community Article

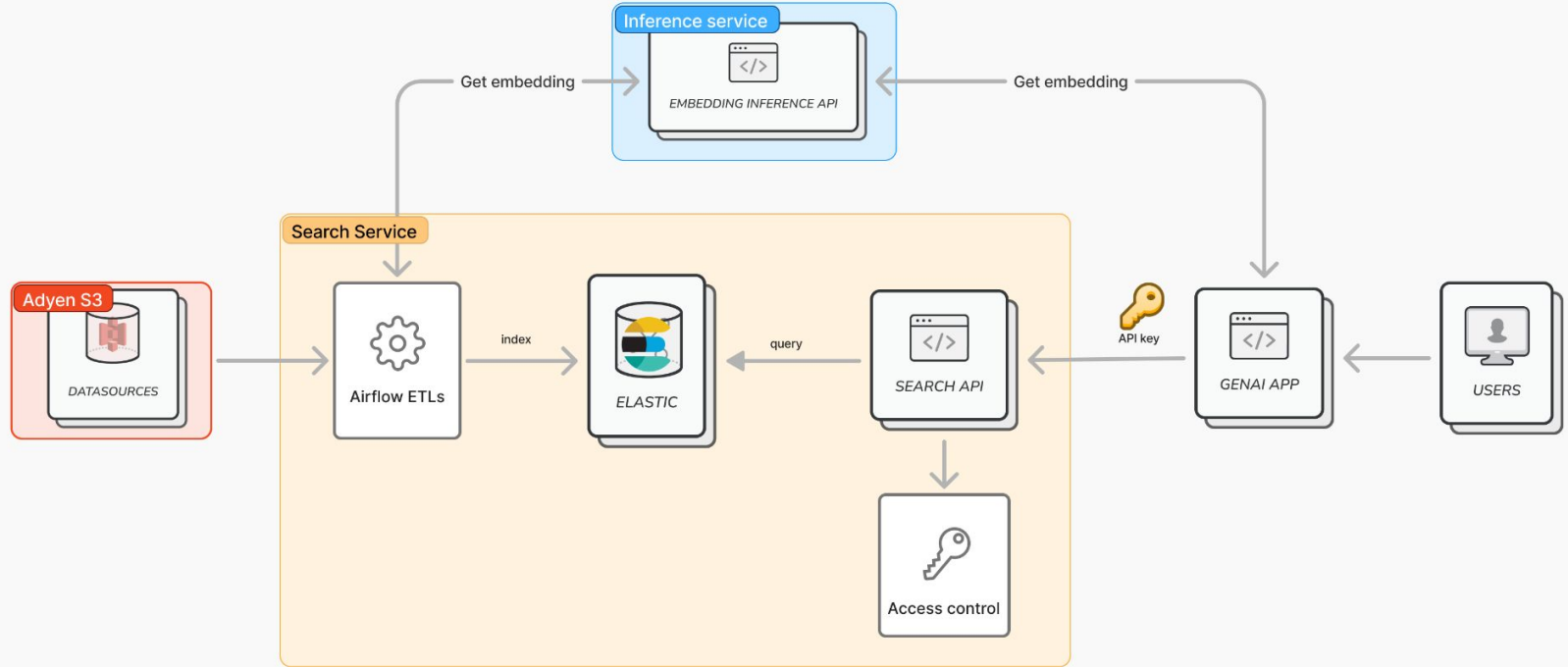
Published September 6, 2024



[martinigoyanes](#)
Martin Iglesias Goyanes

GenAI Platform - Search service

Search service architecture



GenAI Platform - Observability



Langfuse
(APM)

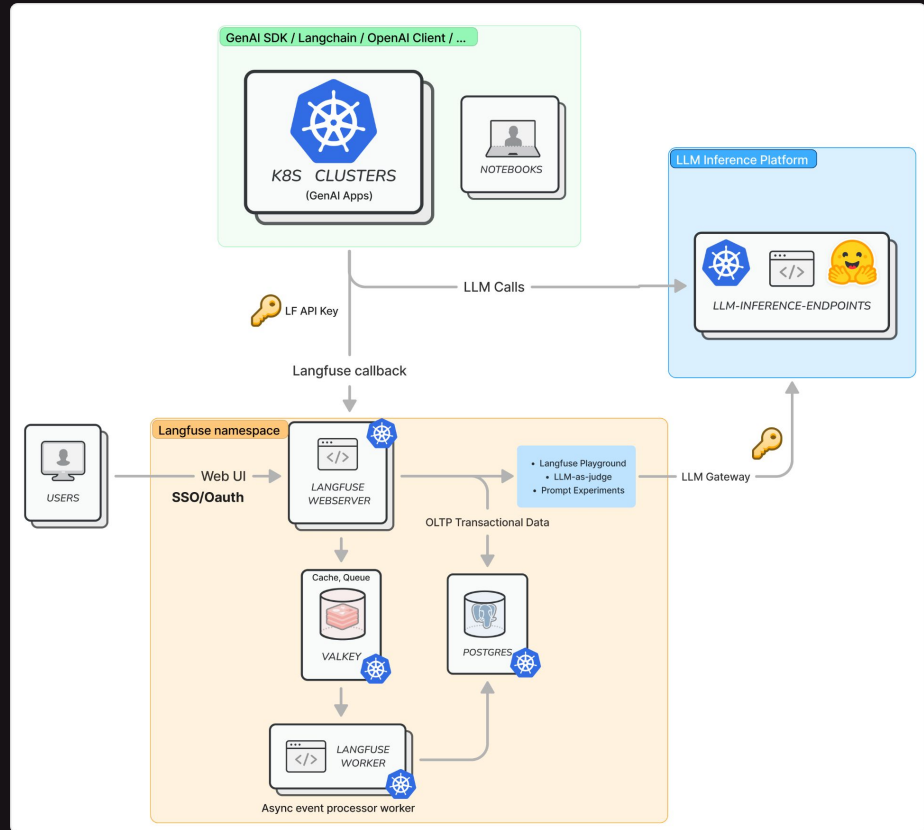


ElasticSearch
(Logs)

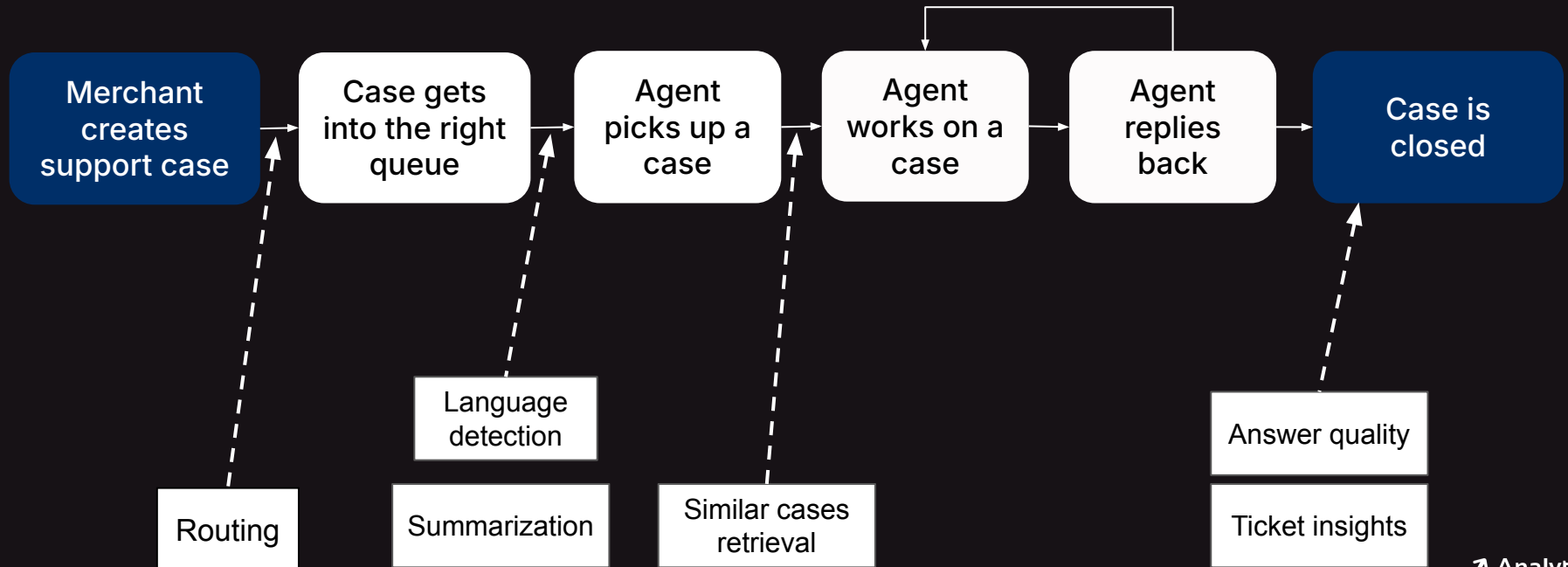


Prometheus
(Metrics)

Observability architecture



GenAI applications (Customer Support)



Real-time CRM integration

CommunicateRedact Case & EmailOmni-ChannelFilesHistoryRelatedAdyen Product

▼ Description

Description

We need assistance in configuring platform alerts to ensure timely notifications of critical events. This includes alert rule settings, event types, and notification channels. Your support is crucial for effective platform monitoring.

▼ AI Details

AI Line Summary ⓘ

Configure platform alerts for critical events

Summarization

Communicate	Redact Case & Email	Omni-Channel	Files	History	Related	Adyen Product
 Case History (6)						
Date	Field	User	Original Value		New Value	
07/02/2025 18:04	Type	Ops AI Integration			Platforms & Financial Services	
07/02/2025 18:04	Topic	Ops AI Integration			Adyen For Platforms	
07/02/2025 18:04	Sub-Type	Ops AI Integration			Platforms	
07/02/2025 18:04	Case Owner	Ops AI Integration	Support Unclassified		Technical Support	
07/02/2025 18:04	Language	Ops AI Integration			English	

Case routing

Language detection

Similar cases

K

Kuma

For your case, Orders with successful then failed captures, I found these relevant cases:

1.Missing webhook for successful capture on [REDACTED]

62% match

[REDACTED]

Opened 27/05/2024

2.Clarification needed on failed capture event and potential retries or indications in webhooks

59% match

[REDACTED]

Opened 13/11/2024

3.Mismatch between successful payments on Adyen portal and Capture webhook responses (Insufficient balance)

58% match

[REDACTED]

Opened 17/05/2024

Were any of these cases relevant for you?

Your feedback will help us improve this workflow.

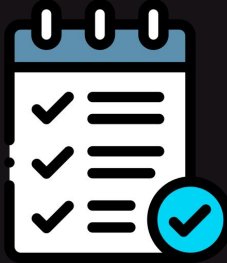
👍

👎

Answer quality



Documentation



Completeness



Personalization



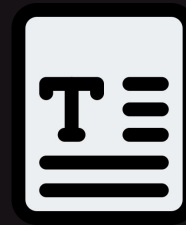
Empathy



Clarity



Accuracy



Formatting

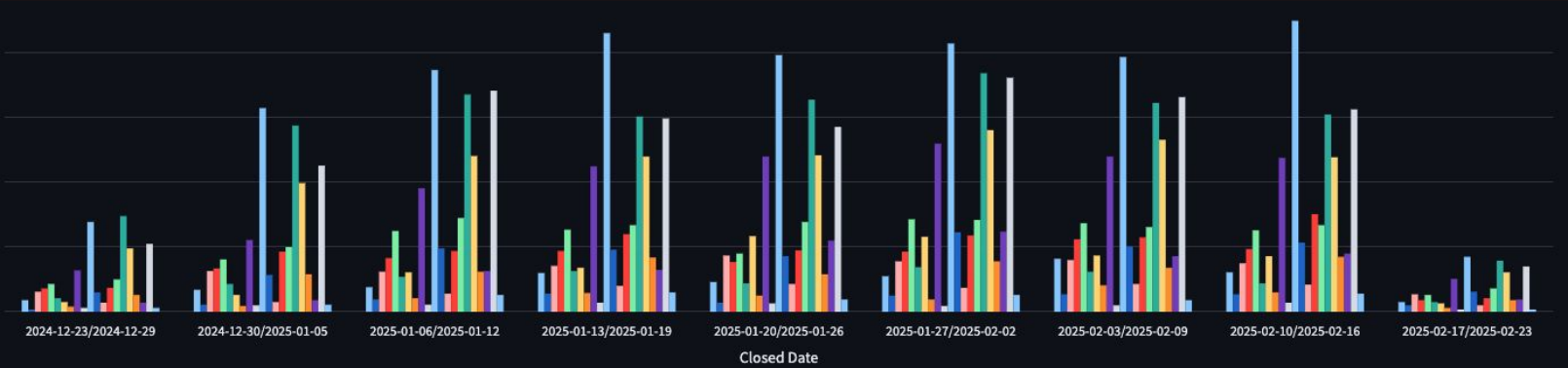


Consistency

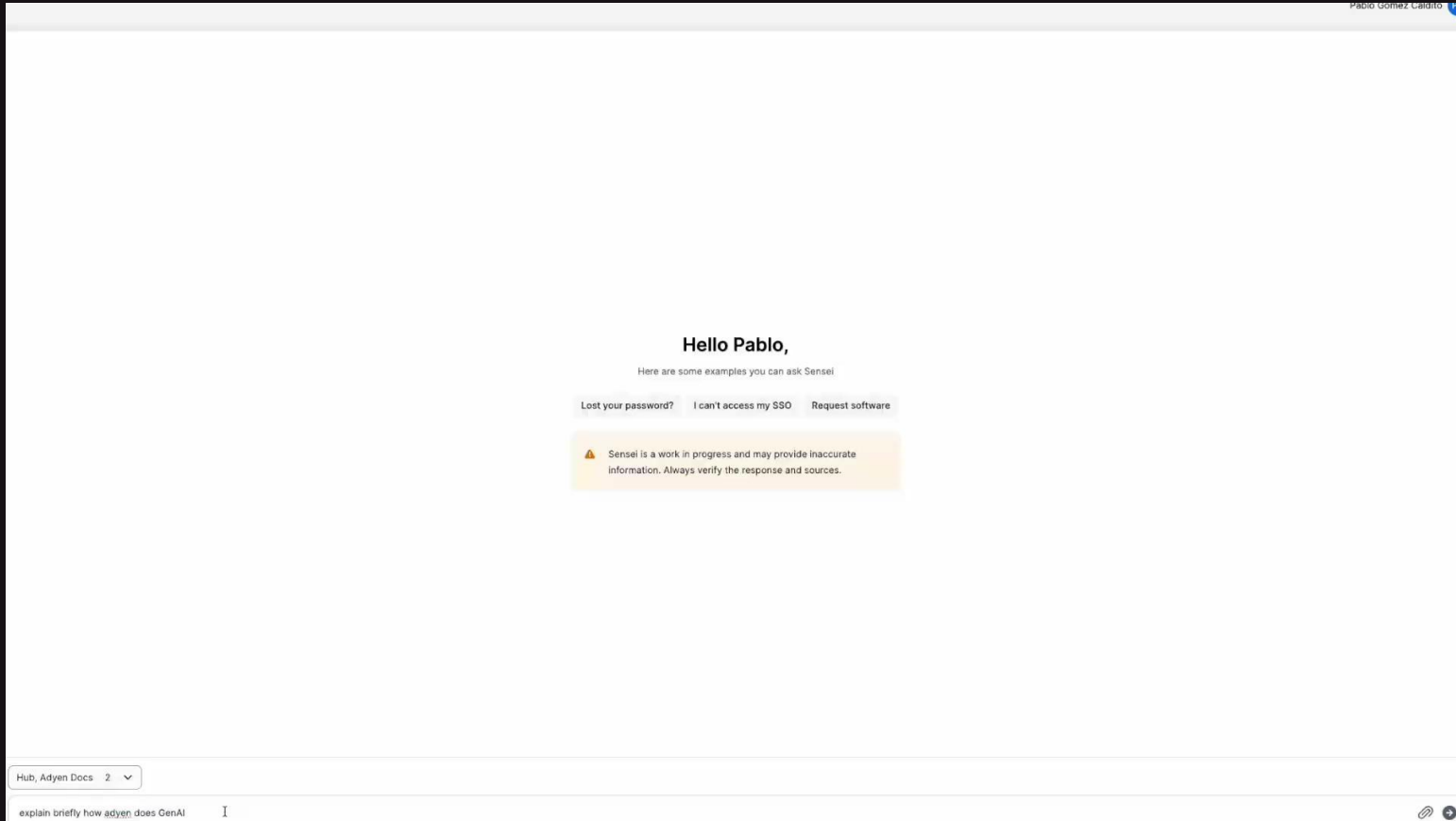
Ticket Insights



case_number	matched_tags	matched_keywords
	payment_method_configuration pre_payment_display_configuration	Payment Methods: Apple Pay
	payment_method_configuration webhook pre_payment_display_configuration	Payment Methods: Klarna Payment Methods: PayPal
	test_environment payment_method_configuration	Payment Methods: PayPal



Chatbot (internal knowledge)



Research (Benchmark) [Link](#)

 Hugging Face [Models](#) [Datasets](#) [Spaces](#) [Posts](#) [Docs](#)



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DABStep: Data Agent Benchmark for Multi-step Reasoning

Published February 4, 2025

[Update on GitHub](#)



[eggie5](#)

Alex Egg

guest



[martinigoyanes](#)

Martin Iglesias Goyanes

guest



[frisokingma](#)

Friso Kingma

guest



[andreumora](#)

Andreu Mora

guest



[lvwerra](#)

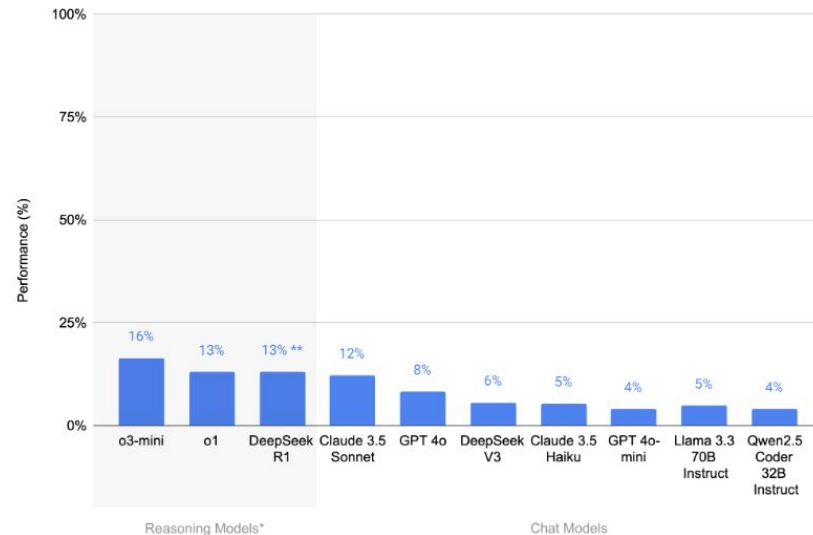
Leandro von Werra



[thomwolf](#)

Thomas Wolf

Baseline Hard

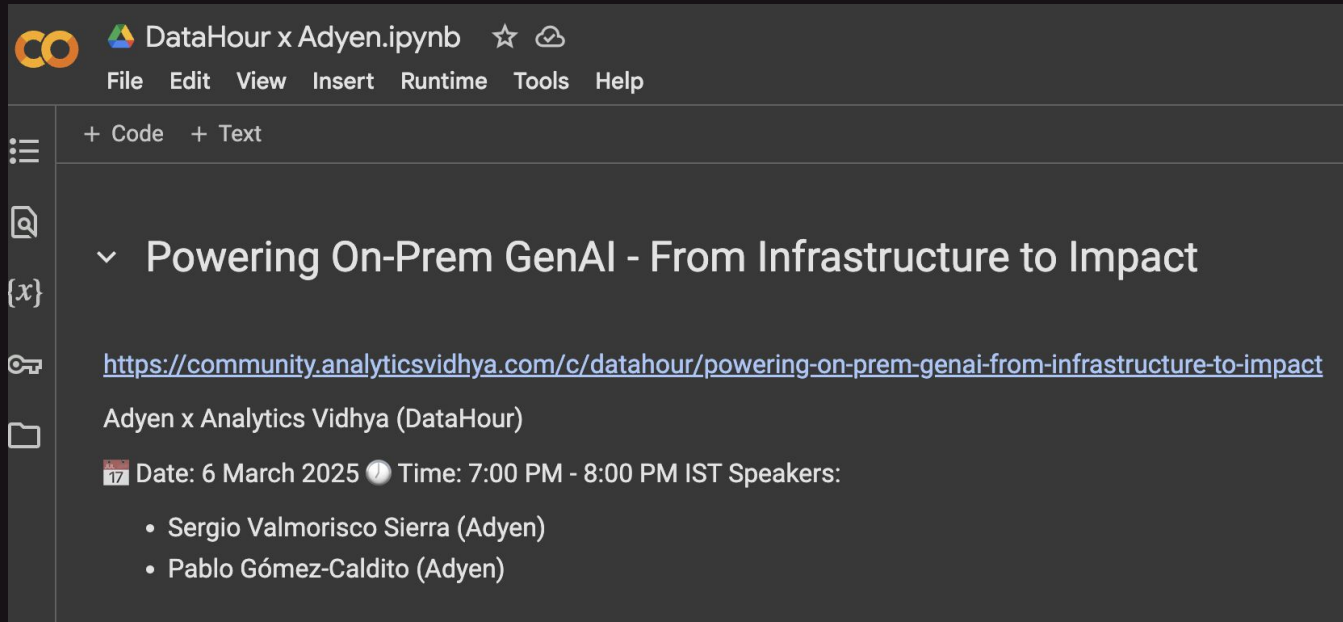


Links

- [Driving Operational Efficiency at Adyen through Large Language Models \(LLMs\)](#)
- [LLMs accelerate Adyen's support team through smart-ticket routing and support agent copilot](#)
- [How Adyen does AI](#)
- [Elevating Code Quality Through LLM Integration](#)
- [LLM inference at scale with TGI](#)
- [DABStep: Data Agent Benchmark for Multi-step Reasoning](#)

Hands-on notebook

[Link](#)



The screenshot shows a Jupyter Notebook interface. At the top, the notebook is titled "DataHour x Adyen.ipynb" with icons for saving, starring, and sharing. Below the title is a menu bar with options: File, Edit, View, Insert, Runtime, Tools, and Help. On the left side, there is a sidebar with icons for a table of contents, search, code input, and file explorer. The main area of the notebook displays the following content:

- A section header: **Powering On-Prem GenAI - From Infrastructure to Impact**
- A URL: <https://community.analyticsvidhya.com/c/datahour/powering-on-prem-genai-from-infrastructure-to-impact>
- Text: Adyen x Analytics Vidhya (DataHour)
- Text:  Date: 6 March 2025  Time: 7:00 PM - 8:00 PM IST Speakers:
- A bulleted list of speakers:
 - Sergio Valmorisco Sierra (Adyen)
 - Pablo Gómez-Caldito (Adyen)

